Chapter 1

Messages on a grid: quadrature, truncation and their diagnostics

The recursion of **??** is exact for functions. A computer holds a function by finitely many numbers, and the paper's choice is the simplest one: the values of each message at N_{g} equally spaced points of an interval $[-A, A]$, with integrals replaced by the composite trapezoidal rule. The paper summarises the consequence in one clause, "exact for the discretised factor model and controlled for the continuous one", and devotes Appendix B to measuring the control. This chapter explains the clause word by word: what the discretised model is and why forward–backward is exact for it; where the trapezoidal rule's error comes from, why it is far smaller than its textbook order for smooth rapidly decaying integrands and why it is not for a discontinuous kernel; what happens when the domain is too narrow for the data; and what each of the paper's diagnostics (column-mass defect, boundary belief mass, drift, exceedance count) measures. It ends with the normalisation bookkeeping of the implementation.

1.1 Functions as vectors

1.1.1 The grid and its weights

Fix a **half-width** $A > 0$ and a number of points $N_{\mathrm{g}} \geq 2$. The **grid** is $u_k = -A + (k-1)h$ for $k = 1, \ldots, N_{\mathrm{g}}$ with **spacing** $h = 2A/(N_{\mathrm{g}} - 1)$; the paper's operating grid has $N_{\mathrm{g}} = 401$, $A = 8$ and hence $h = 0.04$. A function g on $\mathbb{R}$ is represented by the vector $(g(u_1), \ldots, g(u_{N_{\mathrm{g}}}))$, and an integral by the **composite trapezoidal rule**

$$\int_{-A}^{A} g(a)\,\mathrm{d}a \;\approx\; \sum_{k=1}^{N_{\mathrm{g}}} w_k\, g(u_k), \qquad w_1 = w_{N_{\mathrm{g}}} = \tfrac{h}{2}, \quad w_k = h \text{ otherwise.} \tag{1.1}$$

The weights come from integrating the piecewise linear interpolant of g through the grid values: each interior cell $[u_k, u_{k+1}]$ contributes $h\,(g(u_k) + g(u_{k+1}))/2$, and summing over cells gives every interior point the weight h and the two end points half of it. The code is `make_grid` in `src/markov_score/bp_grid.py`.

1.1.2 The discretised factor model

Replacing the integrals of the chain posterior by Equation (1.1) amounts to replacing the continuous model by a finite-state one. Define the **discretised factor model** as the probability mass function on $\{1, \ldots, N_{\mathrm{g}}\}^L$

$$\hat{p}(k_1, \ldots, k_L \mid \mathbf{x}) \propto \mu(u_{k_1}) w_{k_1} \prod_{i=2}^{L} K(u_{k_i} \mid u_{k_{i-1}}) w_{k_i} \prod_{i=1}^{L} \ell_t(x_i \mid u_{k_i}). \tag{1.2}$$

It is a hidden Markov model with N_{g} states whose transition matrix is $\hat{K}[k,l] = K(u_k \mid u_l)\, w_k$, that is, the continuous kernel sampled on the grid and multiplied by the quadrature weight of the destination state. Two remarks. First, $\hat{K}$ is not exactly stochastic: its column sums $\sum_k \hat{K}[k,l]$ are trapezoidal estimates of $\int K(a \mid u_l)\,\mathrm{d}a = 1$ and miss one by the quadrature and truncation error of that integral. The recursion normalises every message, so this deficit never accumulates, but it is the natural scalar diagnostic of the discretisation at its source and is measured as the **column-mass defect** $\max_l \left|\sum_k K(u_k \mid u_l) w_k - 1\right|$ (Section 1.4). Second, running the two sweeps of **??** with the matrix $\hat{K}$ is the forward–backward algorithm on this finite-state model, and by the exactness theorem of that chapter it returns the marginals and the normaliser of Equation (1.2) *exactly*, up to floating point. The returned evidence is $\log \hat{Z}(\mathbf{x})$, the normaliser of the discretised model: exact bookkeeping of an approximate object. This is the first half of the paper's clause. The tests `test_xi_matches_brute_force_enumeration` in `tests/test_em_bp.py` and `tests/test_fisher_bruteforce.py` check exactly this statement, by enumerating Equation (1.2) on a tiny grid.

The second half, "controlled for the continuous one", concerns the distance between the marginals of Equation (1.2), read as functions on the grid, and those of the continuous posterior. That distance has two sources, treated in turn: **quadrature error**, from replacing integrals over $[-A, A]$ by sums, and **truncation error**, from ignoring the part of $\mathbb{R}$ outside the domain.

1.2 The trapezoidal rule and its error

1.2.1 Euler–Maclaurin

Let g be $2m$ times continuously differentiable on $[-A, A]$ and write $T_h(g)$ for the right-hand side of Equation (1.1). The **Euler–Maclaurin formula** states

$$\begin{aligned} T_h(g) - \int_{-A}^{A} g\,\mathrm{d}a &= \sum_{j=1}^{m-1} \frac{B_{2j}}{(2j)!} h^{2j} \Big[g^{(2j-1)}(A) - g^{(2j-1)}(-A) \Big] + R_m, \\ |R_m| &\le C_m\, h^{2m} \int_{-A}^{A} |g^{(2m)}|\,\mathrm{d}a, \end{aligned} \tag{1.3}$$

where $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$, … are the Bernoulli numbers and C_m is an absolute constant. The derivation for one cell is worth seeing once. On $[0, h]$, integrate by parts with the polynomial $P_1(a) = a - h/2$ (whose derivative is 1 and which is odd about the midpoint): $\int_0^h g\,\mathrm{d}a = [P_1 g]_0^h - \int_0^h P_1 g'\,\mathrm{d}a = \frac{h}{2}(g(0) + g(h)) - \int_0^h P_1 g'\,\mathrm{d}a$. The first term is the trapezoidal rule on the cell; the second is the error. Integrating by parts again with P_2 chosen so that $P_2' = P_1$ and $P_2(0) = P_2(h) = h^2/12$ turns it into $-\frac{h^2}{12}(g'(h) - g'(0)) + \int_0^h P_2 g''\,\mathrm{d}a$, and summing over the cells telescopes the boundary terms to the two ends of $[-A, A]$, giving the $j = 1$ term of Equation (1.3) with $B_2/2! = 1/12$. Continuing the integrations by parts generates the higher Bernoulli terms. The textbook conclusion is that the trapezoidal rule is **second order**: its error is $O(h^2)$, with constant $\frac{1}{12}|g'(A) - g'(-A)|$. The paper's Appendix B calls the quadrature error "second order in the spacing" for this reason.

1.2.2 Why the rule is much better than second order on a Gaussian

Look again at Equation (1.3). Every term of the sum involves derivatives of g at the end points of the domain. If g and all its derivatives are negligible there, as they are for a Gaussian-like integrand whose domain is several standard deviations wide, every power of h in the expansion has a vanishing coefficient, and the error is smaller than any power of h. This property is called **spectral accuracy** (or super-algebraic convergence). It can be made exact for an integrand on

the whole line: for g smooth and rapidly decaying, the **Poisson summation formula** gives

$$h\sum_{k\in\mathbb{Z}} g(kh) - \int_{\mathbb{R}} g\,\mathrm{d}a = \sum_{n\neq 0} \hat{g}\Big(\frac{2\pi n}{h}\Big), \qquad \hat{g}(\omega) = \int_{\mathbb{R}} g(a)e^{-i\omega a}\,\mathrm{d}a, \tag{1.4}$$

so the error of the infinite trapezoidal sum is the sum of the Fourier transform of g at the frequencies $2\pi n/h$: it decays as fast as $\hat{g}$ does. For the standard Gaussian density, $\hat{g}(\omega) = e^{-\omega^2/2}$ and the error is $2\sum_{n\geq 1} e^{-2\pi^2 n^2/h^2}$, dominated by $2e^{-2\pi^2/h^2}$. At $h = 1$ this is $2e^{-19.74} \approx 5\times 10^{-9}$; at $h = 0.5$ it is $2e^{-79} \approx 10^{-34}$, below double precision; at the operating $h = 0.04$ it is zero for every practical purpose. The worked example of Section 1.6 checks the $h = 1$ figure on a finite domain.

This is why the quadrature is not the binding error in the paper's numerics for smooth families, and why the paper writes that "the domain, not the spacing, is what a heavy-tailed family tests". Two situations degrade the picture. If the integrand is narrow compared with h, which happens to the unary factor $\ell_t(x_i \mid \cdot)$ at small t (its standard deviation as a function of a is $\sqrt{\Delta_t}/e^{-t}$, about 0.32 at $t = 0.05$, eight cells at $h = 0.04$), the Fourier transform decays slowly and the terms of Equation (1.4) are no longer negligible; the code exposes `likelihood_resolution_ok` in `src/markov_score/noising.py` for this check, and the paper's E04 refines the grid on the reported cells to show the effect is small. And if g is not smooth the expansion Equation (1.3) fails at the first derivative that does not exist: the Laplace kernel has a kink at $a' = \rho a$ (its first derivative jumps), which costs the Bernoulli terms beyond $j = 1$ but still leaves an $O(h^2)$ error; the uniform kernel is discontinuous, and that is the case of the next subsection.

1.2.3 A discontinuous kernel is first order

The uniform innovation has kernel $K(a' \mid a) = 1/(2h_u)$ for $|a' - \rho a| \leq h_u$ and zero otherwise, with $h_u = \sqrt{3q}$ so that the variance is q. As a function of the integration variable it is a step, and the trapezoidal rule on a step is only **first order**. The mechanism is elementary. If a jump of height J falls at a distance $\delta \in [0, h)$ from the nearest grid point, the piecewise linear interpolant misplaces the jump by up to h and the integral by an amount between $-Jh/2$ and $+Jh/2$, depending on δ; the error oscillates with the position of the jump and its envelope shrinks only like h, not h^2. Section 1.6 exhibits the error exactly in the aligned case. In the chain the jumps sit at $a' = \rho a \pm h_u$ and move with a, so every column of $\hat{K}$ has its own misplacement, and the paper's Appendix B records the signature: for the uniform family the worst drift is largest on the *coarsest* spacing rather than on the narrowest domain, the opposite of the heavy-tailed families. The column-mass defect of the uniform kernel is likewise the largest among the families at moderate correlation, and it does fall with N_{g}.

1.3 Truncation

1.3.1 A site beyond the domain

The second error is the one the paper's sweep found to be binding. The grid represents only $[-A, A]$, so every message is implicitly zero outside. For a family whose marginal has light tails (Gaussian, Laplace, the bimodal mixture, the generalised Gaussian) the probability that any site of any sequence lies beyond $A = 8$ is negligible: for a standard Gaussian site it is about 10^{-15}, and for the Laplace innovation at the frozen correlation the probability that a single innovation exceeds 8 in absolute value is of order 10^{-10}. For a Student-t innovation the tail is polynomial and the probability is not negligible, which is the case treated next.

Suppose a clean site has $|a_i| > A$. Its observation is $x_i = e^{-t}a_i + \sqrt{\Delta_t}z_i$, and the unary factor $\ell_t(x_i \mid a)$, as a function of the latent a, is a Gaussian centred at $x_i/e^{-t} \approx a_i$, outside the grid. Restricted to the grid the factor is a monotone ramp, largest at the boundary point nearest to a_i; the transition factors from the neighbouring sites add little against it. The belief

at site i is therefore concentrated in the last few cells at the edge: the paper's phrase is that the posterior is **pinned to the boundary**. The posterior mean is then close to $\pm A$ instead of close to a_i, and by Tweedie's identity the score error at that site is about $(e^{-t}/\Delta_t)\,(|a_i| - A)$, which at the lowest level of the schedule is a factor $e^{-t}/\Delta_t \approx 10$ times the excess. One such site in a bundle of thousands can dominate a pooled relative error, and, crucially, it is pinned on *every* grid with the same half-width, so refining N_{g} at fixed A does not move it. That is the signature the paper's Appendix B reports for the binding cell: drift flat in N_{g}, falling only when the domain widens.

1.3.2 The exceedance count and its Poisson law

The right summary of truncation for a polynomial tail is not a maximum over sites but a count. Let N_A be the number of sites among the ML of a bundle with $|a_i| > A$. Its expectation is

$$\mathbb{E}N_A = ML\,\mathbb{P}(|a| > A) \tag{1.5}$$

when every site has the same marginal law (at $\rho = 0$ every site after the first is an independent draw of the innovation). Because the events are rare and, along a chain, only weakly dependent, the count is close to a Poisson variable with that mean: this is the law of rare events (Poisson's limit theorem, or Le Cam's inequality for dependent indicators with small total variation). No finite A makes Equation (1.5) zero for a polynomial tail; widening the domain only lowers the mean, and a Poisson variable with mean λ is zero with probability $e^{-\lambda}$ and at least one with probability $1 - e^{-\lambda}$. For the Student-t innovation with ν degrees of freedom, variance-matched so that $a = s\,T_\nu$ with $s^2 = q(\nu - 2)/\nu$, the tail probability is $\mathbb{P}(|a| > A) = 2\,\bar{F}_\nu(A/s)$ with $\bar{F}_\nu$ the survival function of T_ν; it decays like $A^{-\nu}$. Section 1.6 computes the mean count for the paper's grids. The paper's Appendix B states this law and gives the expected counts at $A = 12$ and 16; the point is that the Student-t cell of the family ladder is affected by a handful of sites, whose number is predicted, and not by "the family" in some diffuse sense.

1.4 The diagnostics

The paper measures four quantities, all produced by `experiments/e11_grid_validation.py` and the diagnostic branch of `grid_bp_batch`. For each the definition, what it detects, and what it does not.

Column-mass defect. $\max_l \left|\sum_k K(u_k \mid u_l) w_k - 1\right|$: the worst deviation of a column of $\hat{K}$ from unit mass. It combines two effects. In the interior of the grid it is the quadrature error of a one-dimensional integral of the kernel, spectrally small for a smooth kernel (machine precision for the Gaussian and mixture families at the operating grid) and $O(h)$ for the uniform one. In the columns near the boundary it is a *truncation* effect: the column with parent $u_l = \pm A$ is a density centred at $\pm\rho A$, and the part of it beyond $\pm A$ is cut off. As ρ rises, ρA moves towards the edge and the cut-off mass grows, from nothing at $\rho = 0$ to a sizeable fraction of a column at $\rho = 0.95$; this part is flat in N_{g} and falls only with A. A reader checking the paper's Appendix B should therefore read the growth of the defect with correlation as a statement about the outermost columns, not about resolution in the interior, where the smooth families are resolved to machine precision at every correlation. In either case the defect is not an error bar for the posterior: the recursion renormalises every message, so a defect that is the same across a column's neighbours cancels in the normalised beliefs, and the paper reports it as a diagnostic only.

Boundary belief mass. For each site of each chain, the mass of the normalised belief (or of the forward message) in the outermost 2% of the grid on each side, $\sum_{k \le e} b_i(u_k) w_k + \sum_{k > N_{\mathrm{g}} - e} b_i(u_k) w_k$ with $e = \lfloor 0.02\, N_{\mathrm{g}} \rfloor$, maximised over sites and chains. It is computed inside

the recursion, on the very objects the recursion propagates, which is the reason the diagnostic lives in `grid_bp_batch` rather than in a separate tool: an earlier version computed a "boundary mass" from the likelihood table alone, which involved no message and measured nothing the algorithm used. A value near one means that some site's posterior is pinned to the boundary, and the count of such sites is the exceedance count above. A value of a few per cent at $\rho = 0.85$ with no site beyond the domain is a different, benign situation: a site near $|a| \approx 6$ whose posterior at the lowest level has a tail reaching the edge. The paper quotes the frozen configuration's value and the binding cell's value in Appendix B.

Drift. The relative root-mean-square difference between the posterior-mean field computed on a candidate grid and on the finest grid of the ladder, on the same chains and noise draws, pooled over the twelve levels (and separately at the worst level and at the lowest level). Drift is what a reported number would move by under refinement, and it is the quantity a claim is judged against; the other diagnostics say why it is small. Its reference is the finest grid, not the truth, so it is a self-convergence diagnostic: a site pinned on both grids contributes no drift, which is why the binding cell's reference grid in E11 could itself miss the largest site and why E18 and the wide-domain reruns E07b and E07c widen the reference further.

Exceedance count. Not computed by the sweep but by Equation (1.5) from the family's tail; it is the interpretation of the previous three for heavy tails, and it was the audit's re-reading of the E11 cell that turned "all the belief mass outside the domain" into "one or two sites of eight thousand".

1.4.1 How the experiments use them

E11 (the envelope). Five families, four correlations, twelve levels, four seeds, seven candidate grids against a reference at a finer and wider grid. Its findings are the paper's Appendix B: at the frozen configuration the drift is far below every reported effect; across the sweep the worst drift is flat in N_{g} at fixed A and falls only when A grows (truncation, in the Student-t cell at $\rho = 0$); the column-mass defect grows with correlation but does not predict the drift; the uniform family is the one case whose worst drift sits on the coarsest spacing. *E11b* repeats the sweep at chain lengths 64 and 128 to show that the envelope measured at $L = 32$ covers the longer chains of the length study; its prediction is that per-site errors compound at most linearly in L. *E07b and E07c* recompute the family ladder of the paper's Appendix F on wider domains ($A = 12$ and 16), which by Equation (1.5) lowers the expected exceedance count by an order of magnitude and then by another. *E18* recomputes the closure-gap panel of the paper's Figure 2 at $(N_{\mathrm{g}}, A) = (1601, 16)$ and $(3201, 32)$ and shows that the Student-t departure from the e^{-8t} law on the operating grid is the truncated prior's tail in the reference, not the moment condition; on the wide domain the family reaches the predicted constant. *E04* re-scores the validation-selected fitted kernels of the capacity experiment at $N_{\mathrm{g}} = 801$, 1601 and at $A = 10$, with the reference recomputed at each resolution, to bound the movement of the reported estimands under refinement; the paper's Appendix B quotes the bound and notes it is orders of magnitude below the effects.

1.5 Normalisation and the evidence on the grid

The implementation in `src/markov_score/bp_grid.py` keeps messages in the linear domain and rescales them after every step, accumulating the logarithms of the scale factors. The details matter for reading the code and for trusting the returned log-evidence.

The likelihood table. The unary factors are tabulated with the row maximum removed,

$$\log \ell_t(x_i \mid u_k) \text{ is stored as } -\frac{(x_i - e^{-t}u_k)^2}{2\Delta_t} + c_i, \qquad c_i = \max_k \frac{(x_i - e^{-t}u_k)^2}{2\Delta_t},$$

so that the largest entry of every row is exactly 1 after exponentiation and no row can underflow to zero (`log_likelihood_matrix` in `src/markov_score/noising.py`; `grid_bp_batch` recomputes the same table internally). The Gaussian constant $-\frac{1}{2}\log(2\pi\Delta_t)$ is dropped at the same time. Both removals are harmless for the beliefs, which are normalised, and both are recorded: the sum of the row shifts and L copies of the constant are added back to the evidence at the end.

The sweeps. With $\hat{K}$ the matrix $K(u_k \mid u_l)$ (weights applied on the incoming side) the forward step is $\overrightarrow{m}_{i+1} = \hat{K}\,(\overrightarrow{m}_i \circ \ell_i \circ w)/c_{i+1}$, where $c_{i+1} = \sum_k w_k[\hat{K}(\overrightarrow{m}_i \circ \ell_i \circ w)]_k$ is the trapezoidal mass of the unnormalised result; the backward step is the same with $\hat{K}^\top$ and its own constants, which are not needed for the evidence. Each step is a matrix–vector product, or, for a batch of B chains, a matrix–matrix product; the batch is processed in chunks of at most 256 chains so that the $(L \times B \times N_\mathrm{g})$ temporaries fit in memory, with results identical to round-off across chunk sizes (`tests/test_bp_grid_chunk.py`). A guard raises an error if any normaliser is non-finite or non-positive: a message that has lost all its mass is a failure to report, not to hide.

The evidence. Putting the pieces together,

$$\log \hat{Z}(\mathbf{x}) = \sum_{i=2}^{L} \log c_i + \log \sum_k w_k\, \overrightarrow{m}_L(u_k)\, \ell_L(u_k) + \sum_{i=1}^{L} \bigl(-c_i^{\mathrm{shift}}\bigr) - \frac{L}{2}\log(2\pi\Delta_t), \tag{1.6}$$

which is the log-normaliser of Equation (1.2) to floating point. Its correctness is checked in three independent ways: against the closed-form Gaussian marginal likelihood; against an enumeration of the discretised posterior; and against a Monte Carlo estimate on the Laplace chain, where no closed form exists. The first and the third are tests in `tests/test_em_bp.py`; the second is the file `tests/test_fisher_bruteforce.py`. Its role in the paper is twofold: it is the objective that expectation–maximisation increases (**??**), and, summed over a held-out one-view bundle and divided by the number of transitions, it is the oracle-free selection rule of E20.

Diagnostic outputs. When called with `boundary_frac`, `grid_bp_batch` also returns a dictionary with the maximum and mean boundary mass of the forward messages and of the beliefs, and the number of edge cells used. These are the numbers behind the paper's Appendix B and behind the frozen configuration's docstring in `experiments/frozen_config.py`, which is careful to say that a maximum over sampled sites is a statistic and not a bound.

1.6 Worked examples

The Gaussian, spectral accuracy. Integrate the standard Gaussian density on $[-A, A]$ with the trapezoidal rule. At $A = 8$ and $N_\mathrm{g} = 17$, so $h = 1$, the nine non-negligible grid values are $\phi(0) = 0.398942$, $\phi(\pm1) = 0.241971$, $\phi(\pm2) = 0.053991$, $\phi(\pm3) = 0.004432$, $\phi(\pm4) = 0.000134$, and the sum $h\sum_k \phi(k)$ (the end weights are irrelevant at this precision) is 1.0000000054: an error of 5×10^{-9}, which is $2e^{-2\pi^2}$ to two digits, as Equation (1.4) predicts. Halving the spacing to $h = 0.5$ gives an error below 10^{-15}. By contrast, at $h = 2$ the sum is 1.0144: the Poisson term $2e^{-2\pi^2/4} = 0.0144$ again. The convergence is not h^2; it is exponential in $1/h^2$.

The step, first order. Integrate the uniform density $\frac{1}{2}\mathbf{1}\{|a| \le 1\}$ on $[-2, 2]$ with $N_g = 9$ points, $h = 0.5$. The grid values at $a = \pm 1$ are $\frac{1}{2}$ (the jumps sit on grid points), at $|a| < 1$ they are $\frac{1}{2}$, elsewhere zero; the trapezoidal sum is $h \cdot \frac{1}{2} \cdot 5 = 1.25$, an error of $0.25 = h/2$. At $h = 0.25$ the error is $0.125 = h/2$; at $h = 0.125$ it is 0.0625. The error halves when h halves: first order, with constant $J/2$ per jump, exactly as Section 1.2.3 says. Shifting the grid so that the jumps fall between grid points changes the sign and the size of the error but not its order.

The column-mass defect on a coarse grid. Take the Gaussian kernel $K(a' \mid a) = \mathcal{N}(\rho a, q)$ with $\rho = 0.5$, $q = 0.75$ on the five-point grid $\{-2, -1, 0, 1, 2\}$, $h = 1$. The column with parent $u_l = 0$ is the density $\mathcal{N}(0, 0.75)$ sampled at the five points, (0.0320, 0.2365, 0.4607, 0.2365, 0.0320), and its trapezoidal mass with end weights $\frac{1}{2}$ is 0.9657: a defect of 0.034, almost all of it the tail of a $\mathcal{N}(0, 0.75)$ beyond ± 2, which is 0.021, plus the quadrature error of $h = 1$ on a Gaussian of standard deviation 0.87. The column with parent $u_l = 2$ is centred at 1 and loses the whole of its tail beyond 2; its mass is 0.848, a defect of 0.152 that no refinement of h removes and that a wider domain does. Widening to $[-4, 4]$ at the same spacing brings the worst column to 0.983. This is the mechanism behind the growth of the paper's defect with correlation: the extreme column's centre ρA approaches the edge.

The exceedance count. For the variance-matched Student-t_5 innovation, $s^2 = 0.6$ and $\mathbb{P}(|a| > A) = 2\bar{F}_5(A/\sqrt{0.6})$. At $A = 8$ the argument is 10.33 and the two-sided tail is 1.46×10^{-4}; over $ML = 256 \times 32 = 8192$ sites the expected count is 1.2, so the probability of no exceedance in a bundle is $e^{-1.2} \approx 0.30$ and of two or more is 0.34. At $A = 12$ the expected count is 0.17 and at $A = 16$ it is 0.04. These are the analytic figures the paper's Appendix B quotes next to the observed counts of the E11 cell; a reader can reproduce them from the t survival function alone.

Where it is used. The paper, Section 2 ("messages are functions on $\mathbb{R}$; we carry them on a grid ... exact for the discretised factor model and controlled for the continuous one") and Appendix B (the three levels of exactness, the validation at the frozen configuration, the envelope, the refinement, the floors); Appendix F (the Student-t cell excluded pending a wider domain); Appendix E (E18, the wide-domain closure panel). Experiments E11, E11b, E07b, E07c, E18 and E04. Code: the grid and the two sweeps live in `src/markov_score/bp_grid.py` (`make_grid`, `grid_bp`, and `grid_bp_batch` with its `chunk` and `boundary_frac` arguments); the row-shifted likelihood table and `likelihood_resolution_ok` in `src/markov_score/noising.py`; the drivers are `experiments/e11_grid_validation.py`, `experiments/e18_closure_wide.py`, `experiments/e04_refinement.py` and `experiments/e07_phase_diagram.py`, with the macros emitted by `tools/make_grid_envelope.py`. Tests: for the grid itself `tests/test_grid_convergence.py` and `tests/test_bp_grid_chunk.py`; for the evidence, `tests/test_em_bp.py` and `tests/test_fisher_bruteforce.py`.

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